

Chapter Summary: Capacitors

Physics – Chapter 16

1 Core Definitions

- **Capacitor:** A passive component consisting of two conducting plates (armatures) separated by an insulator (dielectric). It stores energy in an electrostatic field.
- **Capacitance (C):** The ability to store charge. It is the ratio of charge Q to voltage U .
- **Dielectric:** An insulator that increases capacitance by a factor of ϵ_r (relative permittivity).
- **Breakdown Voltage (U_{max}):** The maximum voltage before the dielectric fails (arcs).

2 The Mathematical Toolbox

Fundamental Relationships

- **Charge-Voltage Law:** $Q = C \cdot U$ (Units: C in Farads [F], Q in [Coulombs], U in [Volts])
- **Geometric Capacitance:** $C = \epsilon_r \epsilon_0 \frac{S}{d}$ (S : area [m^2], d : distance [m])
- **Vacuum Permittivity:** $\epsilon_0 \approx 8.85 \times 10^{-12} F \cdot m^{-1}$
- **Stored Energy (W):** $W = \frac{1}{2}CU^2 = \frac{1}{2}QU = \frac{1}{2}\frac{Q^2}{C}$ (Units: Joules [J])

3 Circuit Dynamics: RC Circuits

The speed of charging or discharging is governed by the **Time Constant (τ)**:

$$\tau = R \cdot C \quad (\text{Seconds})$$

Feature	Charging Phase	Discharging Phase
Initial ($t = 0$)	$u_C = 0$ (Short Circuit)	$u_C = E$ (Source)
At $t = \tau$	$u_C = 0.63E$ (63% Charged)	$u_C = 0.37E$ (37% Left)
Steady State ($t \geq 5\tau$)	$u_C \approx E, i \approx 0$ (Open Circuit)	$u_C \approx 0, i \approx 0$ (Discharged)

4 Grouping Capacitors

- **Parallel:** $C_{eq} = C_1 + C_2 + \dots$ (Increases area S ; voltage is same).
- **Series:** $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$ (Increases distance d ; charge is same).

5 Unit Conversions

- $1 \mu F = 10^{-6} F$ (Micro) | $1 nF = 10^{-9} F$ (Nano) | $1 pF = 10^{-12} F$ (Pico)
- $1 cm^2 = 10^{-4} m^2$ | $1 dm^2 = 10^{-2} m^2$

